INTERIM AMENDMENT

MILITARY SPECIFICATION

TRUCKS, LIFT, FORK, GASOLINE;

GENERAL SPECIFICATIONS FOR

This Interim Amendment is issued for use by the U. S. Army Mobility Equipment Command with Military Specification MIL-T-21870, dated 24 May 1966. This Interim Amendment includes the provisions of but does not supersede Amendment 1, dated 26 August 1970.

Page 2

2.1, under SPECIFICATIONS, Military, delete all reference to "MIL-B-11040", "MIL-T-16165, "MIL-M-21861", and "MIL-E-55301"; delete title of MIL-M-3971 and substitute "Meters, Time Totalizing, Non-Hermetically Sealed, Electrical: General Specification for."

Under STANDARDS, Military, delete all reference to "MIL-STD-826".

2.1, under SPECIFICATIONS, Military, add following new reference in correct sequence:

"MIL-V-173 - Varnish, Moisture-and-Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Equipment).

MIL-P-514 - Plates, Identification, Instruction and Marking, Blank.

MIL-F-3859 - Grease Guns, Hand, High Pressure, Lever-Operated; Cartridge and Bulk Loading (14 Ounce and 21 Ounce Capacity)."

Under STANDARDS, Military, add following new reference in correct sequence:

"MIL-STD-209 - Slinging Eyes and Attachments for Lifting and Tying Down Military Equipment.

MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.

MIL-STD-1400 - Engines, Gasoline or Diesel, Methods of Test.

MIL-STD-1472 - Human Engineering Design Criteria For Military Systems, Equipment and Facilities."

FSC 3930

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:39Z
Check the source to verify that this is the current version before use.

Page 3

2.1, under STANDARDS, Military, add following new reference:
"MS53063 - Indicator, Air Cleaner, Intake; Restriction Mechanical Type."

2.2, under Other publications, delete address of The American Society of Mechanical Engineers and substitute: "345 East 47th Street, New York, New York 10017."

Page 4

2.2, under Other publications, add the following new references:
"INDUSTRIAL FASTENER INSTITUTE

Standard 114 - Standard for Break Mandrel Blind Rivets.

(Application for copies should be addressed to the Industrial Fastener Institute, 1505 East Ohio Building, Cleveland, Ohio 44114.)

DEPARTMENT OF HEALTH, EDUCATION, AND WELFARE

45 CFR, Part 85 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)"

3.1, line 2, after first sentence, insert: "The individual truck requirements shall be as specified herein and in accordance with the applicable specification sheet."; line 10, delete: "in components to affect interchangeability".

3.1.1, delete and substitute:

"3.1.1 First article (preproduction model). The supplier shall furnish one or more forklifts for examination and testing within the time frame specified to prove prior to starting production that his production methods and choice of design detail will produce forklifts that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3). The approved preproduction model may be presented for acceptance as the last unit for delivery under the contract after being rehabilitated to a new appearance. This shall include furnishing of new tires."

Source: https://assist.dla.mil -- Downloaded: 20260410-19T10:39Z
Check the source to verify that this is the current version before use.

After 3.1.1, add new paragraph as follows:
"3.1.1.1 Initial production. When specified, the supplier shall furnish one or more forklifts for inspection as specified in 4.3.3.4."

3.1.2.1, delete and substitute:
"3.1.2.1 Castings. Castings provided on all trucks shall be at least of the quality normally used in commercial practice, free from injurious blowholes, porosity, shrinkage, cracks, or other defects. The use of repaired castings in which the repairs could adversely affect the strength or intended use of the castings shall not be permitted. Sufficient material shall be furnished to permit machining where necessary for leak-free sealing at covers, closure points, and fasteners."

Page 5

3.1.3, add at end of paragraph: "There shall be no exposed bolts, clamps, gages, fittings, lifting parts, or appendages that can be caught or hooked while the truck is working in a confined space. Hose reel, if furnished, is excluded from this requirement."

3.1.3.4 and 3.1.3.5, delete and substitute:
"3.1.3.4 Lubrication. All surfaces requiring lubrication shall be provided with a means for lubricating.

3.1.3.4.1 Lubricants. The forklift shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.1.3.5 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.1.3.5.1 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts."

3.1.3.7, line 8, at end of second sentence, add: "Dimensions of hand access openings shall be in accordance with MIL-STD-1472 for bare hand or light clothing."; lines 11 and 12, delete "Drain plugs .....complete drainage," and substitute "Drain holes with removable drain plugs shall provide for complete drainage without disconnecting hose(s)."; add at end of paragraph: "The hydraulic system reservoir, transmission housing, and engine shall be equipped with dipsticks to determine the fluid level. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

dipsticks shall be graduated to show the amount required for filling. Dipsticks shall be accessible without the use of handtools. Oil level filler caps shall be painted orange-yellow in accordance with FED. STD. No. 595, color 13538."

Page 6

3.2.1, at end of paragraph, add: "The engine shall be capable of operating with unleaded gasoline (with or without phosphorous) at any conditions specified herein without any detrimental effects to the engine. When specified (see 6.2), all engines furnished under this specification shall be certified to conform to the current Department of Health, Education and Welfare Standards for Exhaust Emissions as applicable to engines manufactured after January 1, 1970."

3.2.1.1, delete and substitute:

"3.2.1.1 Power and speed rating. When a Qualified Products List (QPL) engine is furnished, the power and speed rating of the engine shall be such that operation of the truck under any of the operational requirements and conditions specified herein will not require horsepower in excess of the intermittent horsepower rating of the engine for the applicable governed speed as established by QPL-11275. When a commercial industrial engine is furnished, the power and speed rating of the engine shall be such that it shall provide sufficient power and torque to operate the truck under any of the operational requirements and conditions specified herein without exceeding the engine manufacturer's published intermittent horsepower rating, or 90 percent of the engine manufacturer's published maximum horsepower rating (whichever is lower) at the applicable speeds."

3.2.1.2: Delete.

Pages 7 and 8

3.2.2.1, second sentence, delete and substitute: "The choke control shall be located on the instrument panel, frame, or body and shall be visible to and within reach of the operator when he is in his normal operating position."

3.2.2.3 and 3.2.2.3.1, delete and substitute:

"3.2.2.3 Alternator. An alternator shall be provided. It shall have output characteristics matched to the electrical load required by the equipment as furnished, plus the additional reserve capacity to support the complete electrical system and accessories furnished with the truck, and a 25 percent safety factor capacity. The alternator shall not utilize externally mounted selenium rectifiers. The alternator shall be driven by single or multiple V-belts conforming to SAE J636 to provide full alternator output at all engine speeds without belt slippage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

3.2.2.3.1 Alternator regulator. An alternator regulator shall be supplied if required by the alternator and electrical system design. Provision shall be made within the electrical system for:

(a) Alternator voltage regulation.
(b) Current limiting to protect alternator enclosed diodes or the alternator if applicable.
(c) Disconnection of circuit to the alternator rotor when the ignition switch is off.
(d) Actuation of ammeter or charge indicator lamps where applicable.
(e) Field current relay switch to prevent rotor current from passing through the ignition switch where applicable."

3.2.2.4, line 3, between the first and second sentences, add: " A device shall be furnished to automatically disconnect the starter motor when the engine is running."; line 8, delete "at 0° F." and substitute "at minus 25° F."

3.2.2.5, line 5, between "padlocking." and "The" add: "There shall be no obstruction above the fuel tank filler neck to prohibit access to the filler pipe with a straight nozzle filler hose."

Page 9

3.2.2.5.2, delete last sentence and substitute: "Dry type replaceable cartridge type air cleaners are permitted and shall be of the automotive type. When dry type air cleaners are furnished, a restriction indicator in accordance with MS53063 shall be furnished and shall be mounted so that it is in full view of the operator when seated in the operator's seat."

3.2.2.6, lines 3 through 6, delete second sentence and substitute: "The cooling system shall maintain an air to boil temperature of greater than 120° F. when tested as specified herein."

3.2.2.6.1, line 3, delete "Specification MIL-B-11040." and substitute "SAE J636."

Page 10

3.2.2.6.4, at end of paragraph add: "The radiator cap shall be painted white in accordance with FED. STD. No. 595, color 17875."

3.2.2.7, delete and substitute:
"3.2.2.7 Governor. The governor shall be adjusted to limit the maximum speed of the truck as specified in 3.6.5. The engine speed shall not exceed the engine manufacturer's recommended continuous duty speed for commercial-industrial engines and shall not exceed the intermittent duty speed as established by QPL 11275 for QPL engines when operated under any condition specified herein."

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:39Z
Check the source to verify that this is the current version before use.

Page 11

3.2.2.10, delete and substitute:

"3.2.2.10 Fungus and moisture resistance. When specified (see 6.2), the electrical circuitry, including all components and connections except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, with 1 percent copper 8-quinolinolate (by weight) based on the nonvolatile content of the varnish:

(a) Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated.
(b) Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

When used, the varnish shall be applied by spray, brush, or a combination of both to give a minimum dry-film thickness of 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish."

3.2.3, lines 13 through 22, delete last four sentences and substitute: "Drivetrain components shall conform to and shall be installed in accordance with each component manufacturer's recommendations. Drivetrain fluid and lubricant temperatures shall not exceed 250° F. and drivetrain universal joints (if used) shall not exceed 180° F. when the forklift truck is tested as specified herein."

Page 12

3.2.3.1.1, lines 4 and 5, delete third sentence and substitute:

"The transmission and torque converter fluid shall be filtered by full-flow filter(s), having replaceable element(s) and having a rating of not greater than 25 micron absolute filtration."

After 3.2.3.1.2, add new paragraph:

"3.2.3.1.3 Hydrostatic transmission. When a power shifted transmission is specified, a hydrostatic transmission may be furnished at the option of the supplier. The hydrostatic transmission shall provide variable speed control of the transmission input to output speed ratio. A removable 10 micron oil filter shall be provided to filter all the oil and shall be installed in the suction line between the reservoir and the transmission. Means shall be provided to check the transmission oil level. The transmission shall provide for positive inching control of the truck (throughout the entire range of engine speed) in the forward and reverse directions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

The inching control shall permit tilting and full lifting speeds with rated load independently of and simultaneously with truck motion while in inching travel. The inching control shall be capable of meeting the requirements of MIL-STD-268 without application of the brakes. A directional control system may be furnished which utilizes a left foot operated device which combines both directional selection and inching functions. When this is provided, the hand operated control specified in 3.4.1.6 is not required. Transmission oil temperature shall not exceed 120° F. above ambient when tested in accordance with MIL-STD-268."

3.2.3.2, delete and substitute:

"3.2.3.2 Torque converter. A torque converter shall be supplied with the power shifted transmission except for hydrostatic type transmission. When a hydrostatic transmission is supplied a torque converter will not be supplied. The torque converter and transmission (except hydrostatic) shall absorb maximum stall operation for a minimum of 30 seconds without damage, permanent deformation, or exceeding the fluid temperature limit of 250° F. with an initial fluid temperature of not less than 157° F. in an ambient of not lower than 30° F."

Pages 12 and 13

3.2.4, delete and substitute:

"3.2.4 Hydraulic system. The system shall have a maximum working pressure (relief valve setting) not in excess of 2,000 psi. Oil for the system shall conform to MIL-L-2104, grade 10. For low temperature operation, oil conforming to MIL-L-10295 may be used. All hydraulic component materials shall be compatible with these oils and shall operate herein with an oil operating temperature range of minus 25° F. to 225° F. The temperature rise in any portion of the hydraulic system shall not exceed 100° F. above ambient during all operations specified herein. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal to or greater than 2 times the maximum working pressure (2 times relief valve setting) without external leakage, damage, or permanent deformation. All hydraulic system components should have a minimum burst pressure that is equal to or greater than 3 times the system relief valve setting. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatter). When a single pump provides the system flow, the hydraulic control valves shall be in series but may be in any order. When a multiple element pump is used, each element shall have all related valves in series but may be in any order. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the truck. The system shall provide for lowering of rated load at a rate of not more than 80 feet per minute in the event

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:39Z
Check the source to verify that this is the current version before use.

of failure of or damage to hydraulic hose supplying the lift cylinder. A 1/4-inch pipe tap test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. Hydraulic components shall provide protection against contamination of hydraulic fluid by direct impingement or seepage of other fluids."

Page 13

3.2.4.1, lines 11 through 14, delete fifth sentence and substitute: "Hose fittings shall be reuseable in accordance with SAE J516. Hose fitting termination shall be 37 degree flare female swivel or hydraulic hose flanged 'O' ring."; at end of paragraph add: "Suction lines and ports shall be sized such that the minimum pump inlet pressure shall be 5 inches of mercury vacuum at an oil temperature of 150° F. plus or minus 10° F. Hydraulic hoses shall be correctly restrained to protect them form unnecessary flexing, pressure surges, and contact with other moving mechanical components."

After 3.2.4.1, add new paragraphs 3.2.4.2 through 3.2.4.7:

"3.2.4.2 Hydraulic pump(s). The hydraulic pump(s) shall withstand a minimum proof pressure of 1-1/2 times the maximum system working pressure without permanent deformation, damage, or external leakage.

3.2.4.3 System filter. The system filter shall be the full-flow type and shall have a replaceable element with a maximum rating of 25 micron nominal. Filter elements with integral housings ('spin-on' type) may be used if desired. A filter shall be located so that all return oil shall pass through the filter. Relief oil may return directly to the reservoir. The filter shall be located so that the filter element may be removed for examination and replacement without removing any components other than the filter cover and without draining the oil from the reservoir. At the maximum system oil flow through the filter and an oil temperature of 150° F. plus or minus 10° F., the pressure drop across a new filter shall not exceed 10 psi. An automatic bypass shall be provided. A removable strainer of 50 mesh opening shall be installed in the suction line between the hydraulic reservoir and the pump.

3.2.4.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The oil capacity of the reservoir shall be not less than 110 percent of the fluid required for the hydraulic system operation. The

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

The inching control shall permit tilting and full lifting speeds with rated load independently of and simultaneously with truck motion while in inching travel. The inching control shall be capable of meeting the requirements of MIL-STD-268 without application of the brakes. A directional control system may be furnished which utilizes a left foot operated device which combines both directional selection and inching functions. When this is provided, the hand operated control specified in 3.4.1.6 is not required. Transmission oil temperature shall not exceed 120° F. above ambient when tested in accordance with MIL-STD-268."

3.2.3.2, delete and substitute:

"3.2.3.2 Torque converter. A torque converter shall be supplied with the power shifted transmission except for hydrostatic type transmission. When a hydrostatic transmission is supplied a torque converter will not be supplied. The torque converter and transmission (except hydrostatic) shall absorb maximum stall operation for a minimum of 30 seconds without damage, permanent deformation, or exceeding the fluid temperature limit of 250° F. with an initial fluid temperature of not less than 157° F. in an ambient of not lower than 30° F."

Pages 12 and 13

3.2.4, delete and substitute:

"3.2.4 Hydraulic system. The system shall have a maximum working pressure (relief valve setting) not in excess of 2,000 psi. Oil for the system shall conform to MIL-L-2104, grade 10. For low temperature operation, oil conforming to MIL-L-10295 may be used. All hydraulic component materials shall be compatible with these oils and shall operate herein with an oil operating temperature range of minus 25° F. to 225° F. The temperature rise in any portion of the hydraulic system shall not exceed 100° F. above ambient during all operations specified herein. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal to or greater than 2 times the maximum working pressure (2 times relief valve setting) without external leakage, damage, or permanent deformation. All hydraulic system components should have a minimum burst pressure that is equal to or greater than 3 times the system relief valve setting. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatter). When a single pump provides the system flow, the hydraulic control valves shall be in series but may be in any order. When a multiple element pump is used, each element shall have all related valves in series but may be in any order. Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the truck. The system shall provide for lowering of rated load at a rate of not more than 80 feet per minute in the event

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

of failure of or damage to hydraulic hose supplying the lift cylinder. A 1/4-inch pipe tap test point shall be located at the outlet of each pump or pressure inlet of each control valve for measuring pressure. Hydraulic components shall provide protection against contamination of hydraulic fluid by direct impingement or seepage of other fluids."

Page 13

3.2.4.1, lines 11 through 14, delete fifth sentence and substitute: "Hose fittings shall be reuseable in accordance with SAE J516. Hose fitting termination shall be 37 degree flare female swivel or hydraulic hose flanged 'O' ring."; at end of paragraph add: "Suction lines and ports shall be sized such that the minimum pump inlet pressure shall be 5 inches of mercury vacuum at an oil temperature of 150° F. plus or minus 10° F. Hydraulic hoses shall be correctly restrained to protect them form unnecessary flexing, pressure surges, and contact with other moving mechanical components."

After 3.2.4.1, add new paragraphs 3.2.4.2 through 3.2.4.7:

"3.2.4.2 Hydraulic pump(s). The hydraulic pump(s) shall withstand a minimum proof pressure of 1-1/2 times the maximum system working pressure without permanent deformation, damage, or external leakage.

3.2.4.3 System filter. The system filter shall be the full-flow type and shall have a replaceable element with a maximum rating of 25 micron nominal. Filter elements with integral housings ('spin-on' type) may be used if desired. A filter shall be located so that all return oil shall pass through the filter. Relief oil may return directly to the reservoir. The filter shall be located so that the filter element may be removed for examination and replacement without removing any components other than the filter cover and without draining the oil from the reservoir. At the maximum system oil flow through the filter and an oil temperature of 150° F. plus or minus 10° F., the pressure drop across a new filter shall not exceed 10 psi. An automatic bypass shall be provided. A removable strainer of 50 mesh opening shall be installed in the suction line between the hydraulic reservoir and the pump.

3.2.4.4 Reservoir. A reservoir shall be furnished to contain the hydraulic system oil. The reservoir shall include a filler opening, screen, cap, and other items required to function as specified herein. The reservoir shall be rigidly mounted to prevent vibrations that could result in leaks, loosening of fasteners, brinelling of material, or cracks. The location shall not impair operational visibility or access to either side of the operator's compartment. The oil capacity of the reservoir shall be not less than 110 percent of the fluid required for the hydraulic system operation. The

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:39Z
Check the source to verify that this is the current version before use.

reservoir shall have sufficient capacity to prevent air entering the system with all hydraulic pistons fully extended and sufficient free air capacity to prevent oil being discharged through the reservoir air vent when maximum return flow of oil is surged into the reservoir from the system. The 'Add' and 'Full' marks on the dipstick shall indicate the correct oil level when the truck is on level ground, the mast is vertical, and the forks are at ground level. The filler cap and tube shall be located to provide for filling from a standard 5-gallon container and to prevent entrance of contaminants throughout filling and checking operations. The reservoir pump suction openings should be located a minimum of 1 inch above the bottom of the reservoir to prevent entrance of settled contaminants. All surfaces inside the reservoir should be treated to prevent corrosion. Except for sealed reservoirs a breather shall be provided and shall filter incoming air. The breather shall preclude the entrance of water induced by rain. The reservoir should be baffled when necessary to provide a tortuous flow path from the reservoir inlet to the reservoir outlet in order to prevent a direct current between the two ports. The reservoir should be equipped with a decelerator tube to limit the return line flow rate to the reservoir to a maximum of 4 feet per second. All return line flow should enter the reservoir at a point which is at least 3 inches below the reservoir minimum 'add' oil level to reduce aeration. The reservoir filler neck shall contain a removable 100 mesh strainer retained in place.

3.2.4.5 Hydraulic control valves. Mounting of the valves shall be rigid and shall permit no vibration which may cause instability of the controls. All control-valve spools shall be spring centered with sufficient force to return the levers to the center position and retain them there throughout all operations. Valve spools exposed to dirt and moisture shall be provided with wipers and seals. Spools shall not bind under all operating conditions specified herein.

3.2.4.6 Cylinders. All hydraulic cylinders shall be pin, socket, or gimbal mounted, and shall be either fabricated from seamless steel tubing, or welded and redrawn tubing. All welding and brazing to the cylinder within the zone of piston operation shall be done prior to final machine honing or rolling of the cylinder. Cylinders shall be so located to provide for ease of maintenance, and replacement. All internal cylinder ports shall be located beyond the area of piston travel. Each cylinder shall have an exclusion device to prevent foreign material and fluids from entering and damaging the seal and other cylinder components. All cylinders shall perform as specified herein without buckling or bending. Each piston shall have a bearing surface concentric with the piston to within 0.005 inch T.I.R. to carry the piston

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:39Z
Check the source to verify that this is the currentversion before use.

side loads against the cylinder walls. The bearing surface shall be a material that will not scar, score, or gall the inside surface of the cylinder. The rod shall be hard chrome plated to prevent corrosion. After plating, maximum rod surface roughness shall not exceed 32 microinches. All cylinders shall withstand a pressure that is equal to 2 times the system relief valve setting without evidence of rupture, permanent deformation, damage, or external leakage.

3.2.4.7 Hose reels. When hydraulic hose reels are used on the mast to maintain hose tension during mast extension, sheave wheels, guide rollers, guide blocks, or other means shall be provided to prevent the hose from becoming entangled around and riding over the reels or reel flanges."

Page 14

3.3.1, line 6, after the second sentence, add: "The width of the truck shall not exceed the dimension specified on the applicable specification sheet."

Page 15

3.3.1.2, at the end of the paragraph, add: "Top and bottom of fork tips shall be rounded to a minimum radius of 1/8 inch."

Page 16

3.3.2.2, line 2, after first sentence, add: "Each wheel shall be mounted on wheel studs or capscrews with self-centering tapered ends or on tapered boltholes which shall provide a positive method of locating the wheel on the hub."; lines 5 through 9, delete fourth and fifth sentences and substitute: "Tire chain clearances for the drive wheels shall be not less than the clearances specified in SAE J683."

3.3.2.2.1, delete and substitute:
"3.3.2.2.1 Pneumatic tires. The tires shall be new and unused and shall be selected from the sizes listed on MS16968. The tires shall be pneumatic, high pressure, tube type or tubless furnished with nondirectional tread. Tire loading shall not exceed the values shown on MS16968 when tested in accordance with MIL-STD-268."

3.3.2.1.1, delete paragraph number and substitute: "3.3.2.2.2"; at the beginning of first sentence, delete "Tubes" and substitute "When tube type tires are furnished, tubes".

3.3.3, line 3, after first sentence, insert: "There shall be a minimum of 7 inches measured vertically between the lowest point on the steering wheel rim and a line parallel to the floor and passing through the highest point of the unoccupied seat cushion. There shall be a minimum of 4 inches between the rim of the steering wheel and the nearest projection."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

Page 17

3.3.4.1, line 1 from top of page, after "wheel." delete the period and add: "or a combination thereof. Brake backing plate shall be integral with its support or shall be attached and restrained by positive means such as welding, dowel pins, tapered dowel, split ring, or similar method to prevent rotation of the brake backing plate during brake operation. Capscrews and lockwashers installed with or without adhesives and whether tightened into tapped holes or by nuts are not acceptable unless backing plate is also positively restrained."; lines 8 through 12 from top of page delete eighth and ninth sentences.

Page 18

3.3.5.1, delete last two sentences and substitute: "The seat cushion and backrest shall be of foam rubber or polyurethane covered with slip coated vinyl upholstery having a minimum weight of 40 ounces per yard for 54 inch width."

3.3.5.2, line 8, after fifth sentence add: "If finger ring lift latches are provided, the force required to release or fasten the latches shall not exceed 15 pounds."

3.3.5.3, line 5, after "fuel gauge," add: "transmission fluid temperature gauge,"; at end of paragraph, add: "All gauges and switches shall be labeled to describe component and fluid being monitored or function control."

Pages 18 and 19

3.3.5.6, delete and substitute:

"3.3.5.6 Lifting attachments. Permanently affixed attachments that enable the forklift to be lifted in its normal travel position shall be provided. Each attachment shall withstand, without damage to any part of the forklift, 2-1/2 times the resulting vectorial force imposed when the forklift is suspended by the lifting attachments. Attachments for multiple slings shall be located so that: (a) slings will converge at not more than 19 feet above the lowest extremity of the forklift; and (b) not less than 1 inch clearance will be maintained between the slings and the forklift except where a structure (such as an overhead guard) attached to the forklift has sufficient strength to withstand contact with the sling and incurs no permanent deformation. Dimensions and clearances of the lifting attachments shall conform to MIL-STD-209."

Page 19

After 3.3.5.6, add new paragraphs 3.3.5.6.1 and 3.3.5.6.2:

"3.3.5.6.1 Tiedown attachments. Permanently affixed attachments that permit fastening the forklift to the floor or deck of a transportation medium shall be provided. The tiedown attachments shall conform to MIL-STD-209. Attachments shall be rounded and beveled to prevent cutting of the tiedown material.

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:39Z
Check the source to verify that this is the current version before use.

3.3.5.6.2 Railroad transportability. The forklift shall withstand railroad humping of 8 mph in one direction and 8 mph in the opposite direction without permanent deformation to any of its components when tested as specified in MIL-STD-268."

3.3.5.7, delete last sentence and substitute: "No electrolyte shall be furnished."

3.3.5.8, line 2, between first and second sentences add: "Batteries shall be located so they can be cleaned and serviced without removing any component except quick-release battery box cover if battery box is furnished. Battery supports, holddowns, and areas around the installation which could possibly be affected by dripping or seepage of acids shall be protected with acid resistant paints or flexible protective phenolic coatings."

3.4.1, line 4, after first sentence add: "Letter height for control markings shall be equal to or greater than that specified for general dial and panel design for the applicable viewing distance of the 95th percential man, with seat adjusted to rear position, as specified in MIL-STD-1472."

Page 20

3.4.1, at end of paragraph add: "Mechanical push-pull remote controls when furnished shall be of the waterproof antifriction type, conforming to requirements for marine cables in SAE J917, except as specified herein. As a minimum, all exposed parts of the controls shall be fabricated of corrosion-resistant steel. The cables shall be covered with a polyethylene or vinyl jacket. Cable ends shall be tailored to actuate all levers through their respective arcs without binding. Bend radii shall be as recommended by the cable manufacturer plus an increase in radii of 20 percent. The as-installed mechanical efficiency of each installation shall be not less than 71 percent and the no-load friction shall not exceed 5 ounces per foot. The manufacturer's published ratings for continuous operation shall not be exceeded for any required operation of the control specified herein. Forces necessary to actuate direction and hydraulic controls shall be 10 pounds maximum. All control levers shall withstand a minimum force of 100 pounds at the handgrip and shall have sufficient strength to withstand, without permanent deformation or damage, all vibration forces generated during all operations specified herein. The travel of the controls shall not exceed 6 inches from the center position when measured at the end of the grip."

3.4.1.5, at end of paragraph, add: "A manual throttle as specified in MIL-E-11275 is not required."

3.4.1.6, line 3, delete "right hand operation." and substitute " left hand operation."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

Page 21

After 3.4.2.5, add new paragraph 3.4.2.6:

"3.4.2.6 Transmission temperature gauge. A remote indicating electrical transmission oil temperature gauge shall be furnished."

3.5.1, delete and substitute:

"3.5.1 Lights. Each truck shall be equipped with one sealed beam floodlight of not less than 35 watts, mounted on the left side of the mast. Directional adjustment of this light shall be a minimum of 45 degrees above and below horizontal plane. Horizontal adjustment shall be a minimum of 300 degrees to illuminate front and rear areas. When specified (see 6.2), trucks shall be equipped with an identical floodlight mounted on the right side of the mast. Adequate protection against damage by position or guard shall be provided."

3.5.3, line 6, delete "Fork hanger" and substitute "When specified (see 6.2), the fork hanger" ; at the end of the paragraph, add: "The attachment and its components shall show no evidence of permanent deformation when tested as specified in 4.4.3."

Page 22

3.5.4, delete first sentence and substitute: "When specified (see 6.2), a removable closed cab shall be furnished."

3.5.4.1, delete and substitute:

"3.5.4.1 Cab heater. When specified (see 6.2), a cab heater shall be furnished. The cab heater shall be the automotive kind with windshield defroster."

3.5.5, delete first sentence and substitute: "When specified (see 6.2), the sideshift attachment specified in 3.5.3 shall include a load-leveling device to compensate for irregular surfaces that may cause the truck to tilt sidewise."

3.5.7, at beginning of paragraph, add: "An overhead guard shall be furnished."; line 5, delete the period after "tools" and add: ", and no component shall be mounted to the overhead guard."

Page 23

3.6.1, line 2, delete "40 feet per minute" and substitute " 60 feet per minute".

Page 24

3.6.7, delete the first sentence and substitute: "Unless otherwise specified on the applicable specification sheet, when the truck is unloaded, the forward upright tilt shall be 3 degrees plus or minus 1/2 degree, and rearward tilt shall be 10 degrees plus or minus 1/2 degree."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

3.6.8, line 3, delete "those" and substitute "that".

3.6.11, at the end of paragraph, add: "When the truck is unloaded, the underclearance of the truck, other than the mast and the axles, shall be not less than 4 inches. Hydraulic fittings, hoses, tubing, and linkages shall not be the lowest portion of the truck and shall be protected by structural members from striking obstacles."

3.6.12, at beginning of paragraph, add: "The lift assembly shall be capable of holding the rated load at rated maximum lift not less than 2 minutes with not more than 1/2 inch of vertical drift and not more than 1/4 of 1 degree of rotational drift from the vertical. Specified drift rates shall be measured when hydraulic oil temperature at the inlet to each cylinder is not less than 120° F."

Page 25

3.16.3, delete and substitute: "3.6.13".

Page 26

3.6.18, delete last sentence: "When a high strength .... exceeding 3 inches."

3.6.19, at end of paragraph, add: "When accelerating to maximum speed in reverse direction, steer wheels shall maintain contact with the surface on which the truck is operating."

3.7, delete and substitute:
"3.7 Workmanship. The truck shall perform any operation specified herein without permanent deformation, breakage of connections, malfunction, or component interference caused by incorrect workmanship. All parts of the truck before and after painting shall be clean and free from sand, rust, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. Edges and surfaces exposed to operating and maintenance personnel shall be smooth and rounded to the extent that a hazardous surface does not exist."

3.7.1, delete and substitute:
"3.7.1 Rivet connections. Rivet holes, in the sizes recommended by standard practice, shall be accurately formed and shall have burrs removed. Rivets shall be driven with power tools. Rivet heads shall be full, neatly made, concentric with rivet holes, in full contact with the surface of the member, and shall be in accordance with SAE J492 or Industrial Fastener Institute Standard for Break Mandrel Blind Rivets. Excessive upsetting of rivets to fill holes will not be acceptable."

3.7.3, lines 1 and 2, delete "punched, drilled, reamed, or cast" and substitute "formed".

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

Page 27

3.7.3, line 1 at top of page, after "locking device" delete the period and insert: "or method."; after "This device" insert: "or method"; line 4 from top of page, delete "properly tightened." and substitute " correctly torqued and all other threaded fasteners shall have tightening and thread engagement in accordance with SAE J475."

After 3.7.3, add the following new paragraphs:

"3.7.4 Steel and other metal fabrication. Metals used in the fabrication of the truck shall provide original quality surface finish and shall be free from kinks and sharp bends. Metals having corroded or pitted surfaces are not acceptable. The straightening of materials shall be done by methods that will not cause weakening or injury to the material. Burrs and sharp edges in holes and on sheets, plates, and members shall be removed sufficiently to assure correct fits and to prevent loosening of fasteners and damage to components. Flame cutting may be employed instead of shearing or sawing. Splatter shall be removed from exposed cuts and from re-entrant cuts. Heated metals shall be allowed to cool slowly, except in the performance of designed heat treatment, and overheating shall be avoided in accordance with the recommendations of the metal manufacturer. All modular assembly fabrication shall provide for interchangeability of components.

3.7.5 Machine work. All machined parts shall be manufactured to insure accuracy of assembly through the use of correct jigs, fixtures, or tape controlled machines, or any desired combination of these. Like parts shall be interchangeable."

3.8.1, delete and substitute:

"3.8.1 Walkway coating. Floor plates and step surfaces shall be coated or matted with type II or III, color 37038, black, conforming to MIL-W-5044."

3.9, 3.9.1, and 3.9.2, delete and substitute:

"3.9 Electromagnetic interference characteristics. The forklift shall be suppressed in accordance with class III D of table I of MIL-STD-461. In addition, measurements shall be taken from front and rear of the vehicle at 33 foot distance."

Pages 27 and 28

3.10, delete and substitute:

"3.10 Identification marking. The truck shall be identified with an identification plate conforming to MIL-P-514, type I, style 1, composition C, of grade A, class 1 material which shall be in a visible and safe location on the instrument panel or fire wall of the truck. The identifying data to be applied to the identificati on plates shall be as indicated herein and as defined in MIL-STD-130. The truck shall also be equipped with instruction, warning, and caution plates conforming to MIL-P-514, type III, composition C,

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

grade A, class 1 material prominently located and describing any special or important procedure to be followed in operating, loading, and servicing of the truck and its components. All plates shall be securely attached to the truck with screws, bolts, or rivets, and shall be furnished and mounted by the supplier."

Page 28

After 3.10, add new paragraphs 3.10.1 and 3.10.2:

"3.10.1 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be furnished and in addition shall show the silhouette of the forklift in transport position indicating the center of gravity and the location and capacity of the lifting and tiedown attachments. The plate shall be attached by screws, bolts, or rivets in a conspicuous location. Wheel loading information required in 3.10.2 may be combined in one plate with the requirements specified herein.

3.10.2 Wheel loading plate. Each forklift shall be equipped with a wheel loading plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material. The plate shall be attached by screws, bolts, or rivets in a conspicuous location. As a minimum the plate shall have the following information:

Wheel loading (no load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds

Wheel loading (rated load on forks)
Drive wheels (each) pounds
Steer wheels (each) pounds"

3.11, delete and substitute:

"3.11 Vehicle marking. The assigned registration number for each truck shall be painted in black enamel in block numbers 3 inches high on each side of the upright mast. For pneumatic tired trucks the tire pressures for each tire shall be stenciled in black block numbers 1 inch high on each side of the truck near the applicable tire."

3.12, delete.

Add new paragraph 3.13:

"3.13 Sound level. The sound level of the forklift shall be either, (a) 90 decibals or less at the operator's station or (b) not exceed the sound levels listed in table I or table II at the operator's station and a sound level of 90 decibals at 50 feet when tested as specified in 4.4.2."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

Add Tables I and II:

"Table I. Maximum Steady State Noise Level for Army Materiel Command Equipment

(Commercial Frequencies (ASA Z24.10-1953))

Octave Band Limits (cps)	Center Frequency (cps)	Noise Level (db)
37.5 - 75	53	120
75 - 150	106	115
150 - 300	212	109
300 - 600	425	101
600 - 1200	850	93
1200 - 2400	1700	89
2400 - 4800	3400	89
4800 - 9600	6800	91

Table II. Maximum Steady State Noise Level for Army Materiel Command Equipment

(Preferred Frequencies (ASA S1.6-1960))

Octave Band Limits (cps)	Center Frequency (cps)	Noise Level (db)
44 - 87	63	119
87 - 175	125	114
175 - 350	250	107
350 - 700	500	99
700 - 1400	1000	91
1400 - 2800	2000	89
2800 - 5600	4000	89
5600 -11200	8000	91"

Page 29

4.1, delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Source: https://assist.dla.mil -- Downloaded: 20260-19T10:39Z
Check the source to verify that this is the current version before use.

After 4.2, add new paragraph:

"4.2.1 Preproduction inspection. Prior to examination and test of the forklift, the following shall be performed:

(a) Engine performance test (see 4.3.3.2.1.1). Any change made to the sealed settings shall constitute failure of all tests.
(b) Service the forklifts with the oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
(c) Limber up the forklift as prescribed by the supplier.
(d) Set the hydraulic system relief pressure as specified in 3.2.4. Any change made to this setting shall constitute failure of all tests.
(e) Furnish a schedule of maintenance to be followed during all testing of the preproduction forklift. A list of tools required to perform this maintenance shall also be provided."

Pages 29 and 30

4.3.3.2.1, delete the first sentence and substitute: "The preproduction truck shall be tested in accordance with 4.4.1 through 4.4.8 and MIL-STD-268 to determine compliance with the specification. Forks subjected to 300 percent of rated load shall be inspected for cracks 6 inches each side of heel by magnetic particle or dye penetrant method."

Page 30

After 4.3.3.2.1, add new paragraphs:

"4.3.3.2.1.1 Engine performance (QPL engine). Prior to examination and tests of the preproduction truck, the engine to be installed in the preproduction truck shall be tested in accordance with Test Method Series 2000 of MIL-STD-1400 except Test Method 2400 shall not be performed. Nonconformance to 3.2.1 or failure to meet the requirements of Test Method Series 2000 of MIL-STD-1400 shall constitute failure of this test.

4.3.3.2.1.2 Intake manifold vacuum. During tests specified to meet requirements of 3.6.1, 3.6.5, 3.6.6 and 3.6.19, the engine speed (rpm) and intake manifold vacuum shall be recorded throughout the speed range. The data recorded shall be compared with the data recorded during conduct of the test specified in 4.3.3.2.1.1. If at any time during the test the vacuum reading falls below that recorded during Test Method Series 2000, MIL-STD-1400 for the applicable speed it shall be considered as exceeding the QPL rating of the engine and shall constitute failure of the test."

4.3.3.2.2, line 3, at end of first sentence add: "Each production lift cylinder shall be tested as specified in 4.4.7."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

Add Tables I and II:

"Table I. Maximum Steady State Noise Level for Army Materiel Command Equipment

(Commercial Frequencies (ASA Z24.10-1953))

Octave Band Limits (cps)	Center Frequency (cps)	Noise Level (db)
37.5 - 75	53	120
75 - 150	106	115
150 - 300	212	109
300 - 600	425	101
600 - 1200	850	93
1200 - 2400	1700	89
2400 - 4800	3400	89
4800 - 9600	6800	91

Table II. Maximum Steady State Noise Level for Army Materiel Command Equipment

(Preferred Frequencies (ASA S1.6-1960))

Octave Band Limits (cps)	Center Frequency (cps)	Noise Level (db)
44 - 87	63	119
87 - 175	125	114
175 - 350	250	107
350 - 700	500	99
700 - 1400	1000	91
1400 - 2800	2000	89
2800 - 5600	4000	89
5600 -11200	8000	91"

Page 29

4.1, delete and substitute:

"4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

After 4.2, add new paragraph:

"4.2.1 Preproduction inspection. Prior to examination and test of the forklift, the following shall be performed:

(a) Engine performance test (see 4.3.3.2.1.1). Any change made to the sealed settings shall constitute failure of all tests.
(b) Service the forklifts with the oils, greases, and fuel specified herein and designated for use in the ambient temperatures at which the tests will be conducted.
(c) Limber up the forklift as prescribed by the supplier.
(d) Set the hydraulic system relief pressure as specified in 3.2.4. Any change made to this setting shall constitute failure of all tests.
(e) Furnish a schedule of maintenance to be followed during all testing of the preproduction forklift. A list of tools required to perform this maintenance shall also be provided."

Pages 29 and 30

4.3.3.2.1, delete the first sentence and substitute: "The preproduction truck shall be tested in accordance with 4.4.1 through 4.4.8 and MIL-STD-268 to determine compliance with the specification. Forks subjected to 300 percent of rated load shall be inspected for cracks 6 inches each side of heel by magnetic particle or dye penetrant method."

Page 30

After 4.3.3.2.1, add new paragraphs:

"4.3.3.2.1.1 Engine performance (QPL engine). Prior to examination and tests of the preproduction truck, the engine to be installed in the preproduction truck shall be tested in accordance with Test Method Series 2000 of MIL-STD-1400 except Test Method 2400 shall not be performed. Nonconformance to 3.2.1 or failure to meet the requirements of Test Method Series 2000 of MIL-STD-1400 shall constitute failure of this test.

4.3.3.2.1.2 Intake manifold vacuum. During tests specified to meet requirements of 3.6.1, 3.6.5, 3.6.6 and 3.6.19, the engine speed (rpm) and intake manifold vacuum shall be recorded throughout the speed range. The data recorded shall be compared with the data recorded during conduct of the test specified in 4.3.3.2.1.1. If at any time during the test the vacuum reading falls below that recorded during Test Method Series 2000, MIL-STD-1400 for the applicable speed it shall be considered as exceeding the QPL rating of the engine and shall constitute failure of the test."

4.3.3.2.2, line 3, at end of first sentence add: "Each production lift cylinder shall be tested as specified in 4.4.7."

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

4.3.3.2.3: Delete.

After 4.3.3.3, add new paragraphs as follows:

"4.3.3.4 Initial production inspection. When specified (see 3.1.1.1), one or more initial production forklift trucks will be selected at random by the Government from forklift trucks being produced from production tooling and will be examined as specified in 4.3.3.1 and tested as specified in 4.3.3.2.1 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of an initial production forklift truck shall not exclude the remaining forklift trucks from the quality conformance inspection and acceptance provisions specified in section 4.

4.3.3.4.1 Inspection failure. Failure of the initial production forklift truck to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3.3.4 shall be cause for rejection of the initial production forklift truck and shall be cause for refusal by the Government to continue acceptance of production forklift truck until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the supplier at no cost to the Government on forklift trucks previously accepted and produced under the contract. Any deviations found as a result of the initial production test will be considered prima facie evidence that all forklift trucks accepted prior to the completion of the initial production test are similarily deficient unless evidence to the contrary is furnished by the supplier and such evidence is acceptable to the contracting officer.

4.3.3.5 Inspection comparison. The Government may select forklift trucks at any time during the contract production period and subject these trucks to the examination and tests specified in 4.3.3.4 to determine whether the quality of the selected forklift truck is equal to the quality standards established on previously tested proproduction models or initial production trucks. The inspection will be performed by the Government at a site selected by the Government. Forklift trucks will be selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model(s) and initial production trucks.

4.3.3.5.1 Inspection failure. Failure of any forklift truck when inspectec as specified in 4.3.3.5 may be considered cause for refusal by the Government to continue acceptance of forklift truck until objective evidence furnished by the supplier reveals that corrective action has been taken to eliminate the condition which caused the rejection."

Source: https://assist.dla.mil -- Downloaded: 202640-19T10:39Z
Check the source to verify that this is the currentversion before use.

4.3.4, delete and substitute:
"4.3.4 Examination of preparation for delivery. The trucks shall be examined to determine conformance to the preservation, packaging, packing, and marking requirements of MIL-STD-162."

4.4.1, delete and substitute:
"4.4.1 Electromagnetic interference characteristics. The preproduction model equipped for electromagnetic interference in accordance with 3.9 shall be tested as specified in MIL-STD-461. The supplier shall furnish the contracting officer the report of the test required by MIL-STD-461. Upon approval of the report by the contracting officer and provided all other requirements of the specification are met, the preproduction truck shall be used as a model for all production trucks. The report shall include a list of the components used to reduce the electromagnetic interference."

After 4.4.1, add new paragraphs 4.4.2 and 4.4.2.1, 4.4.2.2, 4.4.3, 4.4.4, 4.4.5, 4.4.6, 4.4.7, and 4.4.8:
"4.4.2 Sound level measurement.

4.4.2.1 Required test. Measure the sound level at the operator's station in accordance with test procedures specified in SAE J919. Data collected shall be made with an A-weighted sound level meter set for slow or equivalent meter response. If the sound levels measured for any of the test conditions specified in SAE J919 exceeds 90 decibles, the additional tests specified in 4.4.2.2 shall be conducted.

4.4.2.2 Additional tests. Conduct the following tests at the condition producing the highest A-weighted sound level meter reading measured in 4.4.2.1. Use an octave band analyzer to measure sound levels in accordance with test procedures specified in SAE J919. Measure sound levels at 50 feet in accordance with test procedures specified in SAE J952. Nonconformance to 3.13 shall constitute failure of the test.

4.4.3 Sideshift attachment. When a sideshift attachment is furnished, the following test shall be performed: Operate the sideshift attachment for 100 full cycles at full speed from extreme left to extreme right and back to extreme left per cycle with the rated load on the forks. During the last 50 cycles, the control shall be released abruptly when the load is in approximately mid position between the left and right extremes and then the sideshifter shall be restarted. Upon completion of the 100 cycles, the attachment and its components shall be examined for compliance with 3.5.3. Nonconformance to 3.5.3 shall constitute failure of this test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.

4.4.4 Starter disconnect. Start the vehicle by energizing the starter switch and keep the starter switch energized a minimum of 10 seconds after the engine starts. Release starter switch for a minimum of 5 seconds and reactivate starter switch while engine is operating. Any evidence of starter engagement while engine is running or not disengaging when engine starts shall constitute failure of this test. Repeat this test at conclusion of the endurance test.

4.4.5 Hydraulic pump(s). Operate the pump(s) at 1-1/2 times maximum system working pressure and maximum operating speed for not less than 2 minutes. Check for external leaks at assembled surfaces and the shaft seal. Evidence of permanent deformation, damage, or external leakage shall constitute failure of this test.

4.4.6 System filter. The test of pressure drop at rated system flow may be conducted on the truck or test bench. Pressure gauges shall be installed at the inlet and outlet of the housing. When the filter outlet is directly into the reservoir, the outlet pressure may be considered to be the reservoir pressure. Install a clean filter element and establish maximum flow through the filter. Nonconformance to 3.2.4.3 shall constitute failure of this test.

4.4.7 Hydraulic cylinders. Subject each hydraulic cylinder to a hydrostatic pressure that is equal to or greater than twice the system relief valve pressure setting and hold this pressure for 30 seconds. Nonconformance to 3.2.4.6 shall constitute failure of this test.

4.4.8 Hydraulic system vacuum. Install a vacuum gage into the pump suction line immediately adjacent to the pump or at the pump. Install a thermocouple in the hydraulic reservoir. Run the pump at governed rpm. Measure the pump inlet pressure in inches of mercury vacuum at the specified temperature and speed. Nonconformance to 3.2.4.1 shall constitute failure of this test."

Page 31

6.2, delete subparagraphs (b), (e), (f), (i), (j), (k), and (m); add the following new subparagraphs:

"(n) Time required for submission of the preproduction model (see 3.1.1).

(o) When initial production inspection is required and number of trucks to be furnished, when applicable (see 3.1.1.1).

(p) When certification to HEW exhaust emissions is required (see 3.2.1).

Source: https://assist.dla.mil -- Downloaded: 202610-19T10:39Z
Check the source to verify that this is the current version before use.

(q) When fungus and moisture resistance is required (see 3.2.2.10).
(r) When additional light is required (see 3.5.1).
(s) When sideshift attachment is required (see 3.5.3).
(t) When fork hanger is required to provide for third fork (see 3.5.3).
(u) When removable closed cab is required (see 3.5.4).
(v) When cab heater is required (see 3.5.4.1).
(w) When load-leveling device is required (see 3.5.5).
(x) When fire extinguisher is required (see 3.5.8)."

After 6.2, add new paragraphs:

"6.3 Preproduction model. Any changes or deviations of production forklifts from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish forklifts conforming to this specification.

6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the forklift covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Definitions.

6.5.1 Air to boil temperature. Air to boil temperature is defined as boiling temperature of the pressurized system minus highest measured radiator top tank temperature (coolant from the engine) plus the ambient temperature during test.

6.5.2 Gross vehicle weight. Gross vehicle weight shall include the complete forklift with lubricants and coolants necessary to completely service the forklift for operation, fuel tank full, 175 pound operator, and rated load."

Preparing activity:

Army - ME

Project No. 3930-A336

☆ U. S. GOVERNMENT PRINTING OFFICE: 1974-713-166/1662

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T10:39Z
Check the source to verify that this is the current version before use.